

Evaluation Criteria Summary

Data Understanding

The dataset consists of paired Earth Observation (EO) optical imagery and Synthetic Aperture Radar (SAR) imagery for post-disaster building damage assessment. The primary objective is binary change detection through semantic segmentation of damaged regions. Initial analysis revealed that the dataset contains multimodal data with significant differences in spatial and structural characteristics between EO and SAR imagery. EO images provide rich spectral and texture information, while SAR images contain structural information but also introduce considerable speckle noise and low-contrast regions. A major challenge identified during exploration was severe class imbalance, where background pixels heavily outnumber damaged pixels. The original labels contained four damage categories, which were remapped into binary classes for simplified segmentation. Several samples contained extremely sparse damaged regions, making the learning process highly biased toward majority background predictions. TIFF image handling, multimodal fusion, and noisy SAR patterns also introduced preprocessing and training complexity. These dataset characteristics significantly influenced model design decisions, loss function selection, threshold tuning, and evaluation strategy throughout the project.

Literature Survey

Remote sensing change detection is an important research area used in disaster monitoring, urban planning, infrastructure assessment, and environmental analysis. Traditional change detection methods such as image differencing, thresholding, and statistical comparison relied heavily on handcrafted techniques and often struggled with complex multimodal imagery. With the advancement of deep learning, convolutional

neural networks became the dominant approach for semantic segmentation and change detection tasks due to their ability to learn spatial and semantic representations directly from data. U-Net is one of the most widely used architectures for segmentation because of its encoder-decoder structure and skip connections that preserve fine spatial information. It has demonstrated strong performance in remote sensing applications involving small and sparse target regions. Recent research has increasingly focused on multimodal fusion using Earth Observation (EO) optical imagery and Synthetic Aperture Radar (SAR) imagery. EO imagery provides rich spectral and texture information, while SAR imagery offers structural information and robustness under poor weather conditions and cloud cover. However, EO-SAR fusion introduces significant challenges because SAR imagery contains speckle noise and spatial characteristics that differ substantially from optical imagery. Recent transformer-based architectures such as ChangeFormer and attention-based segmentation networks have shown improved performance in multimodal change detection tasks by capturing long-range dependencies and improving feature interaction. Another major challenge in remote sensing segmentation is severe class imbalance, where damaged regions are significantly fewer than background pixels. To address this issue, researchers commonly use weighted loss functions, Dice Loss, and Focal Loss to improve sensitivity toward sparse target regions. In this project, a U-Net architecture with Weighted BCE and Dice Loss was implemented to balance segmentation performance, computational efficiency, and robustness under severe class imbalance conditions.

Research Thinking

The design decisions in this project were primarily driven by the characteristics and limitations of the dataset. Initial data exploration revealed severe class imbalance, sparse damaged regions, and noisy SAR imagery, which significantly influenced model selection, loss function design, and training strategy. A U-Net architecture with a ResNet34 encoder was selected because it provides strong segmentation performance while remaining computationally efficient for local hardware training. The encoder-

decoder structure and skip connections were particularly useful for preserving spatial details required for pixel-wise damaged region localization.

During experimentation, standard Binary Cross Entropy loss caused the model to collapse toward predicting the majority background class. To address this issue, Weighted BCE combined with Dice Loss was implemented to improve sensitivity toward sparse damaged regions and stabilize optimization. Threshold tuning was additionally performed to improve recall performance and reduce conservative predictions. The project also involved trade-offs between model complexity and computational feasibility, since transformer-based multimodal architectures could potentially improve EO-SAR feature fusion but would require significantly larger computational resources and training time. Overall, the methodology prioritized building a stable and interpretable baseline pipeline while addressing the major practical challenges associated with EO-SAR multimodal segmentation.

Implementation Quality

The project was implemented as a modular and reproducible end-to-end deep learning pipeline using PyTorch. The workflow included TIFF image preprocessing, binary label remapping, multimodal EO-SAR fusion, dataset loading, model training, evaluation, and qualitative visualization. A custom Dataset and DataLoader pipeline was developed to efficiently handle multimodal image pairs and binary segmentation masks. The codebase was organized into preprocessing, training, evaluation, and visualization stages to improve readability and maintainability. Special attention was given to reproducibility and debugging throughout the implementation process. Dataset structure validation, label verification, and mask analysis were performed before model training to identify preprocessing issues and severe class imbalance. Weighted BCE combined with Dice Loss was implemented to improve training stability under sparse positive samples. Evaluation metrics including IoU, Precision, Recall, F1 Score, and confusion matrix

analysis were integrated directly into the pipeline for systematic performance tracking. The implementation was designed to remain computationally efficient while maintaining flexibility for future experimentation with advanced architectures, augmentation strategies, and loss functions.

Results and Analysis

The final model was evaluated using IoU, Precision, Recall, F1 Score, and confusion matrix analysis on both validation and test datasets. Initial experiments using standard Binary Cross Entropy loss resulted in prediction collapse toward the majority background class because of severe dataset imbalance. After implementing Weighted BCE combined with Dice Loss and applying threshold tuning, the model demonstrated improved sensitivity toward damaged regions, although overall segmentation performance remained limited. The final evaluation produced the following results: IoU = 0.0001, Precision = 0.0009, Recall = 0.0001, and F1 Score = 0.0003. The confusion matrix revealed extremely high background prediction dominance with very few true positive detections for damaged regions. Qualitative visualization analysis showed that the model performed relatively better on large and clearly visible damaged structures, while struggling with sparse damaged regions, low-contrast areas, and SAR speckle noise. Failure cases primarily involved missed detections and fragmented predictions caused by severe class imbalance and multimodal feature complexity. Despite limited quantitative performance, the experiments provided valuable insights into the challenges of EO-SAR multimodal segmentation and highlighted the importance of imbalance handling, threshold optimization, and advanced feature fusion strategies.

Future Work

The current U-Net baseline establishes a functional EO-SAR binary change detection pipeline, but several improvements can be explored to improve segmentation

performance and multimodal feature learning. Future work would focus on transformer-based architectures such as ChangeFormer and Attention U-Net to better capture long-range spatial dependencies and improve EO-SAR feature fusion. Since severe class imbalance remained the primary challenge, advanced loss functions including Focal Loss and Tversky Loss would be investigated to improve sensitivity toward sparse damaged regions. Stronger augmentation strategies, multi-scale training, and patch-based learning could further improve generalization and small-object detection performance. Dedicated SAR denoising and preprocessing techniques would also be explored to reduce speckle noise and improve structural feature extraction. Self-supervised pretraining on large remote sensing datasets may additionally improve feature representation quality under limited labeled data conditions. Future experimentation would also include systematic hyperparameter tuning, threshold optimization, and ablation studies to better understand failure cases and improve segmentation robustness in multimodal EO-SAR environments.

Communication

The project report was structured to present the methodology, experimentation process, results, and analysis in a clear, concise, and technically precise manner. Each stage of the workflow, including data preprocessing, multimodal EO-SAR fusion, model training, imbalance handling, evaluation, and qualitative analysis, was explained systematically to ensure reproducibility and technical clarity. Technical decisions such as the use of Weighted BCE combined with Dice Loss, threshold tuning, and U-Net architecture selection were justified based on observed dataset characteristics and experimental outcomes. Evaluation metrics, confusion matrix analysis, and prediction visualizations were included to provide both quantitative and qualitative assessment of model performance. The report also highlighted limitations, failure cases, and future research directions to provide an honest and research-oriented assessment of the implemented approach.